

Mini Project Problem Statements — Index

Course: UE24AM342AA5 — ML System Design and AgentOps
Component: Jackfruit (Level 3), 10 marks | **Team size:** 2–3 students

Fifteen problem statements. Each team works on one. Technical terms used in all of them are defined in 00_GLOSSARY.pdf.

All fifteen share the same 22 requirements — A1–A8 (agent system), L1–L6 (LLMOps), D1–D8 (platform and deployment) — the same demonstration list and the same assessment. Only the domain, the corpus and the functional requirements differ.

The difficulty in each project is agentic by construction. The case in section 4.1 cannot be resolved by a single retrieval pass, because the provision that decides the outcome is a modifier the request never mentions. It requires an agent that searches, establishes that further provisions may apply, searches again, and verifies its own output.

Every corpus is free to download from a regulator, ministry, university or open-source project.

#	Project	Domain
01	Motor Insurance Claim Intake and Triage Service	General Insurance
02	Unauthorised Transaction and Card Dispute Triage Service	Retail Banking
03	Health Insurance Pre-Authorisation Desk	Health Insurance and Hospital Revenue Cycle
04	Air Passenger Compensation Claim Triage Service	Aviation
05	Building Plan Compliance Pre-Check Service	Urban Development and Construction
06	Public Procurement Bid Compliance Screening Service	Government and Public Sector Procurement
07	Export Classification Check Service	Logistics and Trade Compliance
08	Production Incident Triage and Runbook Assistant	Software Operations and Site Reliability Engineering
09	Workplace Compliance Advisory Desk	Human Resources and Employment Law
10	Packaged Food Label Compliance Check Service	Food Manufacturing and Regulatory Affairs
11	Electricity Tariff and Billing Dispute Triage Service	Power Utilities and Regulation
12	Securities Investor Grievance Routing Service	Capital Markets and Investor Protection
13	Environmental Clearance Pre-Screening Service	Environment and Infrastructure Development
14	Pharmacovigilance Adverse Event Intake and Triage Service	Pharmaceuticals and Drug Safety
15	Academic Regulation and Student Petition Triage Service	Higher Education Administration

Retrieval difficulty by project

#	Obstacle to single-pass retrieval
01	A restriction on use located away from the cover it limits
02	A liability table that cannot be applied without definitions held elsewhere
03	Waiting periods listed away from the cover clause they restrict
04	A condition that removes an entitlement, next to the amounts it governs
05	A road-width limit stated separately from the setback tables
06	Two authoritative documents that contradict one another
07	A heading that names the article, in a different chapter from its material
08	Documentation organised by cause, alerts arriving as symptoms
09	A repealed Act held alongside the Code that replaced it

#	Obstacle to single-pass retrieval
10	A claim condition in a different regulation from the labelling rules
11	A mixed-use rule stated away from the category definitions
12	Investment vocabulary pointing to the wrong regulator
13	A General Condition after the table that changes rows within it
14	Everyday descriptions of reactions the insert lists in medical terms
15	A condonation provision stated away from the rule it softens